

Isaiah Flager-Hearan

Belleville, IL 62221

isaiahhearan@gmail.com | <https://www.linkedin.com/in/isaiahflagerhearan>

<https://www.github.com/PapaZay>

EDUCATION

Southern Illinois University Edwardsville

Bachelor of Science, Computer Science

Dec 2024

Edwardsville, Illinois

WORK EXPERIENCE

National Great Rivers Research and Education Center

Jun 2024 - Aug 2024

Software Engineer Intern

Alton, Illinois

- Collaborated with a cross-functional team of 4 developers to build a responsive web application for visualizing real-time water monitoring data across 3 states (Illinois, Missouri, Wisconsin)
- Designed and implemented an optimized PostgreSQL database architecture for real-time water quality visualization and long-term scientific data storage, improving query performance by 20%
- Built a user-friendly data administration tool in Python, enabling non-technical scientists to manage database content independently without SQL knowledge
- Deployed full-stack application on DigitalOcean with scalable infrastructure, ensuring 99.9% uptime for critical environmental monitoring
- Developed responsive frontend using React.js, Next.js, and Tailwind CSS with interactive Leaflet.js maps for enhanced geospatial data visualization
- Integrated into real-time data pipelines for continuous monitoring station updates and scientific reporting workflows

SKILLS

- **Programming Languages:** Python, TypeScript, Java
- **Backend:** FastAPI, Flask, Django, Spring Boot, REST APIs
- **Frontend:** React.js, Next.js, Tailwind CSS
- **Databases:** MySQL, PostgreSQL, MongoDB, TimeScaleDB
- **Cloud & DevOps:** AWS, Google Cloud, Docker, GitHub Actions, Nginx, DigitalOcean, Supabase
- **Monitoring & Tooling:** Git, GitHub, Linux, Sentry

PROJECTS

Polywatch - Polymarket Signal Detector | polywatch.dev

Dec 2025 - Present

- Built real-time data pipeline collecting trading data from 1,600+ Polymarket prediction markets, processing 750K+ time-series snapshots with automated 5-minute collection cycles
- Implemented statistical pattern detection engine using z-score analysis and sliding window algorithms to identify volume spikes and price momentum anomalies with confidence scoring
- Designed FastAPI backend with SQLAlchemy ORM, TimescaleDB hypertables for time-series optimized PostgreSQL storage, Alembic migrations, and signal deduplication logic for efficient historical querying
- Deployed on Google Cloud Platform using Docker containers on Cloud Run for the API, Cloud Run Jobs with Cloud Scheduler for data collection, Compute Engine for TimescaleDB, and Cloud Storage for static frontend hosting
- Built responsive React dashboard with TanStack Query for auto-refreshing signal alerts and market data visualization with Tailwind CSS

Steam Price Tracker | steampricetracker.com

May 2025 - Present

- Built full-stack price tracking platform enabling Steam gamers to monitor prices across thousands of games and receive automated email alerts on discounts via Mailjet integration
- Designed a recommendation service leveraging OpenAI APIs to generate personalized game suggestions based on user preferences, tracked titles, and genre metadata
- Architected secure authentication with hybrid JWT system (Bearer tokens + HttpOnly cookies) and integrated Stripe payment processing for donations with automatic receipt generation
- Deployed production backend on AWS EC2 with Nginx reverse proxy and Cloudflare DNS, achieving 99.9% uptime with automated hourly price syncing via APScheduler background jobs
- Integrated Sentry for full-stack monitoring with error tracking, performance profiling, and distributed tracing across React/TypeScript frontend and FastAPI backend
- Live API: api.steampricetracker.com/docs

Airspace Safety Monitoring System | windborne-delta.vercel.app/

Oct 2025 - Dec 2025

- Developed real-time collision monitoring dashboard tracking 500+ commercial aircraft and weather balloons, automatically categorizing collision risk into CRITICAL (<10km horizontal, <1km vertical), WARNING, and SAFE zones

- Built interactive geospatial visualization using React, TypeScript, and Leaflet with 3D distance calculation algorithms for airspace safety assessment across global airspace
- Architected hybrid cloud infrastructure using Vercel for frontend and Cloudflare Workers for OAuth proxy microservice, implementing token caching and rate limit management to achieve 99.9% uptime
- Designed responsive TailwindCSS dashboard with color-coded threat visualization and real-time safety statistics for airspace awareness